

30. Ans: C

Since the count rate drops significantly when there is a piece of paper, the radiation is stopped by the paper. Therefore it must be α radiation, which has low penetrative power.

The background count rate is 24 min^{-1} .

Hence initial count rate due to source is $532 - 24 = 508 \text{ min}^{-1}$

After two half-lives, count rate due to source = $\frac{1}{4} (508) = 127 \text{ min}^{-1}$

Hence reading = $127 + 24 = 151 \text{ min}^{-1}$

